Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	xxxxxx
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*The system should allow authorized users to securely access the application. If login functionality is implemented, users must authenticate using valid credentials before accessing prediction features.*	**High**
2	Authorization levels	*The system should support different user roles such as Applicant and Administrator. Applicants can submit loan details and view predictions, while Administrators can manage the application and monitor system activities.*	**High**

3	External interfaces	*The application should interact with the trained Machine Learning model through the Flask backend. Future enhancements may include integration with banking APIs or credit score verification services.*	**Medium**
4	Transactions processing	*The system should accept applicant information, validate the input data, process it through the trained Machine Learning model, and display the loan eligibility prediction accurately.*	**HIGH**
5	Reporting	*The application should display the prediction result to the user. Future versions may generate prediction history, reports, dashboards, and downloadable summaries for administrators.*	**Medium**
6	Business rules, etc.	*Loan eligibility should be determined based on applicant attributes such as income, education, employment status, credit history, loan amount, and loan term using the trained Machine Learning model.*	**HIGH**
7	Compliance to laws or regulations	*The system should ensure the confidentiality and privacy of applicant data and follow applicable data protection and security practices while processing user information.*	**Medium**

8	*Other*	*The application should provide a simple, responsive, and user-friendly interface that allows users to easily enter loan details and obtain predictions.* *High*	**LOW**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*The system should process user inputs and generate loan eligibility predictions quickly to provide a smooth user experience.*	*Prediction result displayed within **2–3 seconds** after form submission.*
2	Scalability	*The application should efficiently handle an increasing number of users and support future expansion with larger datasets and additional features.*	*Supports **multiple concurrent users** without significant performance degradation.*
3	Security & Data Privacy	*The system should securely process applicant information, protect sensitive data, and prevent unauthorized access to user information.*	*Applicant data processed securely with **restricted access** and secure communication.*
4	Reliability & Availability	*The application should consistently provide accurate predictions and remain available whenever users access the system.*	***99% system availability** with consistent prediction performance.*
5	Usability & Accessibility	*The application should provide an intuitive, responsive, and easy-to-use interface that is accessible on modern web browsers and devices.*	*User-friendly interface with **responsive design** and easy navigation.*

6	*Other*	*The system should have a modular architecture that allows easy maintenance, model updates, and deployment across different operating systems.*	Code should be **modular, maintainable, and deployable** on Windows, Linux, and macOS.